01 — What Is a Vector: A Physical Quantity with Magnitude and Direction

> **Kleppner:** Ch 1.1~1.4 | **Time:** 60 min
> **Core Question:** Why isn't a single number enough for physics?

Scene: How Do You Describe an Airplane's Location?

"The airplane is flying at 500 km/h."

Can you find the airplane with just this information? **No.** You don't know which direction it's going.

"The airplane is flying at 500 km/h **eastward** from Seoul."

Now you can find it. **Physics needs direction.**

❌ What If You Try With Algebra Alone?

Attempt 1: Express Everything as a Single Number

The airplane is flying northeast at 500 km/h. Express this as a single number — "500"?

- After 1 hour, you can't know the position. Is it 500 km north? 500 km east? Somewhere in between?

Attempt 2: Split Into Two Numbers

Northward 353.6 km/h, eastward 353.6 km/h. Using **two numbers**, you can determine the position.

But this is still inconvenient: the single piece of information "northeast 500" has been scattered into two fragments. And if the direction changes, the numbers change too.

Conclusion: A single number isn't enough. You need a tool that bundles multiple numbers together and handles them as one.

✅ Resolved With Vectors

Vector = Magnitude + Direction Bundled Together

$$\vec{v} = (353.6,\ 353.6)\ \mathrm{km/h}$$

What this notation means: **"Every hour, move 353.6 km east and 353.6 km north."**

The Three Operations on Vectors — Their Physical Meaning

Operation	Formula	Physical Scene
Addition	$\vec{a} + \vec{b}$	"First go by $\vec{a}$, then from there go by $\vec{b}$." — pathfinding
Dot Product	$\vec{a} \cdot \vec{b} = ab\cos\theta$	"How much does $\vec{a}$ contribute in the direction of $\vec{b}$." — **the length of a shadow.** Work is maximized when force and displacement share the same direction.
Cross Product	$\vec{a} \times \vec{b} = ab\sin\theta\,\hat{n}$	"The parallelogram formed by $\vec{a}$ and $\vec{b}$." Direction given by the right-hand rule. — The root of **rotational effects.** The origin of torque and angular momentum.

Scalar vs. Vector

- **Scalar**: A single number suffices. Mass (5 kg), temperature (300 K), time (3 s).
- **Vector**: Needs a direction attached to the number to have meaning. Position, velocity, acceleration, force.
- Why? **Because space is three-dimensional.** Every point in space is described only by

"how far from the origin, and in which direction."

💡 Examples — Thinking With Vectors

Example 1: Airplane on a Windy Day

An airplane flies north at 200 km/h. Wind blows east at 50 km/h. What is the airplane's actual velocity?

$$\vec{v}_{\text{actual}} = \vec{v}_{\text{plane}} + \vec{v}_{\text{wind}} = (0, 200) + (50, 0) = (50, 200)$$

Magnitude: $\sqrt{50^2 + 200^2} = \sqrt{2500 + 40000} = \sqrt{42500} \approx 206.2 \text{ km/h}$
Direction: $\tan^{-1}(50/200) \approx 14.0^\circ$ — tilted about 14° east of north.

Example 2: Resolving a Force Into Components

A 10 N force makes a 30° angle with the x-axis:

$$F_x = 10 \cos 30^\circ = 10 \times \frac{\sqrt{3}}{2} \approx 8.66 \text{ N}$$

$$F_y = 10 \sin 30^\circ = 10 \times \frac{1}{2} = 5.00 \text{ N}$$

Example 3: Net Force From Two Forces

$\vec{F}_1 = (3, 4)$ N, $\vec{F}_2 = (-1, 2)$ N

$$\vec{F}_{\text{net}} = (3 - 1,\ 4 + 2) = (2, 6) \text{ N}$$

Magnitude: $\sqrt{4 + 36} = \sqrt{40} \approx 6.32 \text{ N}$

Example 4: Meaning of the Dot Product — Work

A 5 N force moves an object 10 m at a 60° angle to the force. How much work is done?

$$W = \vec{F} \cdot \vec{s} = 5 \times 10 \times \cos 60^\circ = 50 \times \frac{1}{2} = 25 \text{ J}$$

If force and displacement were aligned (0°): $W = 50 \times 1 = 50\ \mathrm{J}$ — maximum.
If they were perpendicular (90°): $W = 50 \times 0 = 0\ \mathrm{J}$ — no work at all.

The dot product measures "how aligned two directions are."

Example 5: Meaning of the Cross Product — Torque

Apply a 20 N force perpendicular to a wrench at a distance of 0.3 m.

$$\tau = rF\sin 90^\circ = 0.3 \times 20 \times 1 = 6\ \mathrm{N{\cdot}m}$$

If you push along the wrench ($\theta = 0^\circ$): $\tau = 0$ — it doesn't turn at all.
The cross product measures "how much twisting is attempted."

Example 6: Unit Vectors

$\hat{i} = (1,0,0), \hat{j} = (0,1,0), \hat{k} = (0,0,1)$

Any vector can be expressed as a combination of these three: $\vec{v} = 3\hat{i} + 4\hat{j} - 2\hat{k}$

Meaning: "What We Just Did"

Every physical phenomenon in space is described using vectors. Without vectors, you cannot:

- Decompose projectile motion into components
- State the direction of centripetal acceleration in circular motion
- Express the direction of a force

Vectors are the mathematical translation of the simple fact that "space has direction."
Vector operations themselves are algebraic, but the real story unfolds starting next session, when we introduce the **change-in-a-blink of a vector.**

🔧 Basic Drills *(solutions in `solutions/01-solutions.md`)*

1. $\vec{a} = (3, -2, 5), \vec{b} = (-1, 4, 2)$. Find the magnitude of $\vec{a} + \vec{b}$.
2. A 15 N force makes a 45° angle with the x-axis. Find its x and y components.
3. $\vec{F}_1 = (5, 0)\ \mathrm{N}, \vec{F}_2 = (3, 4)\ \mathrm{N}$. Find the magnitude and direction of the net force.
4. $\vec{a} = (2, 3), \vec{b} = (4, -1)$. Find the dot product. Is the angle between them greater or less than 90°?
5. $\vec{r} = (0, 2, 0)\ \mathrm{m}$, $\vec{F} = (3, 0, 0)\ \mathrm{N}$. Find the magnitude and direction of the torque $\vec{\tau} = \vec{r} \times \vec{F}$.

🔥 Advanced Drills *(solutions in `solutions/01-solutions.md`)*

1. Wind blows northwest at 15 m/s. An airplane flies north at 80 m/s. Find the magnitude and direction of the actual velocity.
2. Find a unit vector perpendicular to both $\vec{a} = (2, -1, 3)$ and $\vec{b} = (1, 2, -1)$.
3. A force $\vec{F} = (2, 3)\ \mathrm{N}$ acts through displacement $\vec{s} = (4, 1)\ \mathrm{m}$. How much work is done? How much does this force contribute in the direction $\vec{n} = (-3, 2)\ \mathrm{N}$?
4. A river flows east at 3 m/s. A boat moves north at 4 m/s relative to the water. What is the boat's velocity relative to the ground? If the river is 100 m wide, how many seconds to cross?
5. Find a unit vector in the direction of $\vec{a} = (1, 2, 2)$. Find the angle between this vector and $\vec{b} = (2, -1, 2)$.

🧠 Intuitional Drills *(solutions in `solutions/01-solutions.md`)*

1. Two equal-magnitude forces $\vec{F}_1$ and $\vec{F}_2$ act at an angle of $120°$. The magnitude of each is $10\ \mathrm{N}$. What is the magnitude of the net force?
2. A force $\vec{F} = (6, 8)\ \mathrm{N}$ acts on a box. At what angle should the box be pushed so that only half the maximum possible work is done per meter of displacement?
3. A boat heads straight north at $4\ \mathrm{m/s}$ across a river flowing east at $3\ \mathrm{m/s}$. Without calculating: is the boat's ground speed greater than, less than, or equal to $5\ \mathrm{m/s}$?

4. Three forces $\vec{F}_1, \vec{F}_2, \vec{F}_3$ act on a point and the net force is zero. If $\vec{F}_1 = (5, 0)$ and $\vec{F}_2 = (-3, 4)$, find $\vec{F}_3$ using geometry — no equations, just draw and close the triangle.
5. $\vec{a} \cdot \vec{b} = 0$. What can you say about the angle? $\vec{a} \times \vec{b} = 0$. What can you say?

📖 Reading Physical Symbols

This session introduced symbols that appear throughout mechanics. Here is how to *read* them — not just compute with them.

Symbol	How to Read It	Physical Meaning
$\vec{a}$	"vector a"	A quantity with **direction** attached. Not just a number — an arrow in space.
$\vec{a} + \vec{b}$	"a plus b"	"Go along $\vec{a}$, then from there go along $\vec{b}$." The diagonal of the parallelogram.
$\vec{a} \cdot \vec{b}$	"a dot b"	"How much of $\vec{a}$ points in the direction of $\vec{b}$." The **shadow** length times $
$\vec{a} \times \vec{b}$	"a cross b"	"The area of the parallelogram spanned by $\vec{a}$ and $\vec{b}$, with a direction given by the right-hand rule." Zero means they are parallel.
$	\vec{a}	$
$\hat{i}, \hat{j}, \hat{k}$	"i-hat, j-hat, k-hat"	The three **reference arrows** pointing along the x, y, z axes, each of length 1. Any vector is built from these.
$\hat{n}$	"n-hat"	A unit vector (length 1) in some direction of interest — often the result of a cross product.
$\perp$	"perpendicular to"	Two things are at right angles. Critical for normal forces (always $\perp$ to surfaces).
$\parallel$	"parallel to"	Two things point in the same (or opposite) direction. Friction is

Symbol	How to Read It	Physical Meaning
		always $\parallel$ to the contact surface.

Key habit: Every time you see a vector symbol, picture an arrow in space. Before calculating, ask: "Which way does it point? How long is it?"

Glossary

- **Scalar**: A quantity with magnitude only (mass, temperature, time)
- **Vector**: A quantity with both magnitude and direction (position, velocity, force)
- **Unit vectors** $\hat{i}, \hat{j}, \hat{k}$: Reference direction vectors of length 1
- **Dot product** $\vec{a} \cdot \vec{b}$: The product of magnitudes × directional alignment. Result is a scalar.
- **Cross product** $\vec{a} \times \vec{b}$: The area and perpendicular direction formed by two vectors. Result is a vector.